

Under-Graduate Project

1 Physical and Mathematical Formulation

In Lagrangian mechanics, a system's state is defined by $L(q, \dot{q}) = T(q, \dot{q}) - U(q)$. Introducing conjugate momenta $p_k = \frac{\partial L}{\partial \dot{q}_k}(q, \dot{q})$, we define the Hamiltonian representing total energy as $H(p, q) = T(p, q) + U(q)$. This yields Hamilton's Canonical Equations:

$$\dot{p}_k = -\frac{\partial H}{\partial q_k}(p, q), \quad \dot{q}_k = \frac{\partial H}{\partial p_k}(p, q)$$

For a normalized mathematical pendulum ($m = \ell = g = 1$), the Hamiltonian is:

$$H(p, q) = \frac{p^2}{2} - \cos q$$

Resulting in the first-order system:

$$\dot{p} = -\sin q, \quad \dot{q} = p$$

2 Numerical Implementation

Using the 4th-Order Runge-Kutta (RK4) method for $\dot{y} = f(t, y)$ with step size h :

$$\begin{aligned} k_1 &= f(t_n, y_n) \\ k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right) \\ k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right) \\ k_4 &= f(t_n + h, y_n + hk_3) \\ y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \end{aligned}$$

3 Computational Results

3.1 Short-Time Trajectory

Trajectories for step sizes $h \in \{2^{-2}, 2^{-3}, 2^{-4}, 2^{-5}\}$ demonstrate the high $\mathcal{O}(h^4)$ consistency of the RK4 method over short intervals.

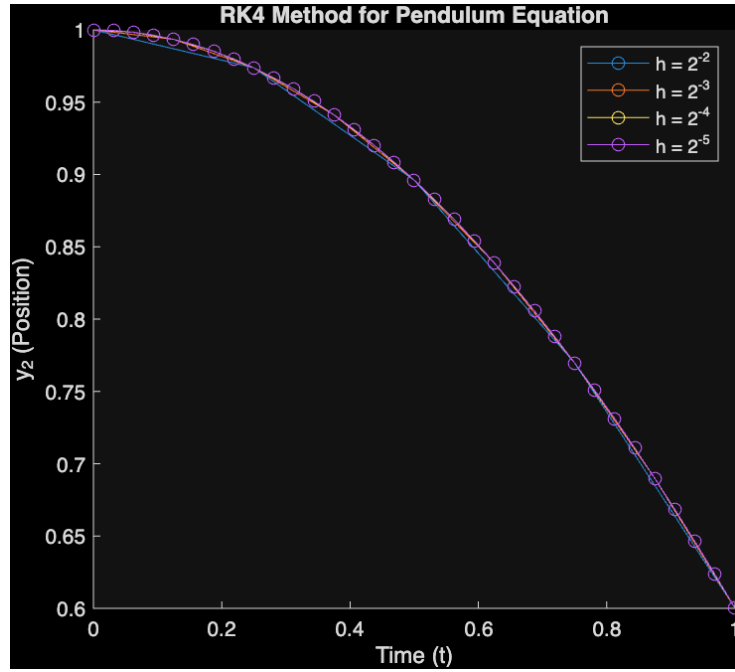


Figure 1: Short-time trajectory of the pendulum.

3.2 Long-Time Trajectory

Simulated up to $t = 100$, the amplitude appears bounded between $[-1, 1]$ with no visually obvious damping.

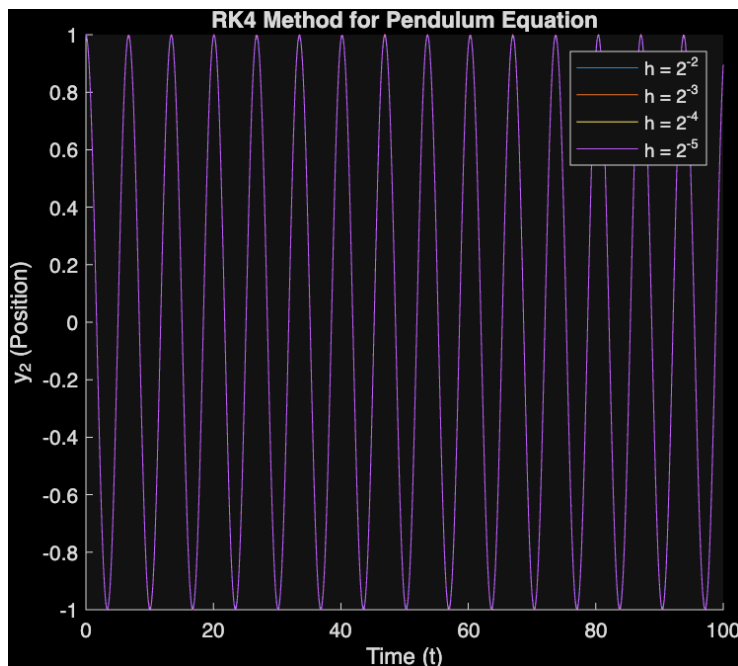


Figure 2: Long-time trajectory showing smooth oscillations.

4 Physical Interpretation and Symplectic Motivation

Classical RK4 is an explicit, non-symplectic integrator. It does not preserve the phase-space area, leading to an invisible but systematic drift in the total energy over extended periods. This limitation motivates the use of Symplectic Methods (which preserve symplectic structure) and Relaxation Methods (which enforce exact energy conservation) for long-term integrations.

5 From Lagrangian to Hamiltonian Mechanics

The evolution of many physical systems is governed by Hamilton's principle, which dictates that the motion follows the Euler-Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = \frac{\partial L}{\partial q} \quad (1)$$

where $L(q, \dot{q}) = T - U$ is the Lagrangian. While powerful, this formulation inherently yields a system of second-order differential equations.

To transform this into a more symmetric system of first-order differential equations, we introduce a new set of variables called conjugate momenta, defined as:

$$p_k = \frac{\partial L}{\partial \dot{q}_k} \quad (2)$$

Using a Legendre transform, we can then define the Hamiltonian function $H(p, q)$ as:

$$H(p, q) = p^T \dot{q} - L(q, \dot{q}) \quad (3)$$

Theorem 1.3. *Lagrange's equations are equivalent to Hamilton's canonical equations:*

$$\dot{p}_k = -\frac{\partial H}{\partial q_k}, \quad \dot{q}_k = \frac{\partial H}{\partial p_k} \quad (4)$$

Proof. By definition, the Hamiltonian is a function of p and q , meaning we must treat $\dot{q}$ as a function of p and q . When we take the partial derivatives of H with respect to p and q using the product rule and chain rule, we obtain:

$$\frac{\partial H}{\partial p} = \dot{q}^T + p^T \frac{\partial \dot{q}}{\partial p} - \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial p} = \dot{q}^T \quad (5)$$

$$\frac{\partial H}{\partial q} = p^T \frac{\partial \dot{q}}{\partial q} - \frac{\partial L}{\partial q} - \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial q} = -\frac{\partial L}{\partial q} \quad (6)$$

Notice how the terms $p^T \frac{\partial \dot{q}}{\partial p}$ and $\frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial p}$ perfectly cancel each other out precisely because we defined our momenta as $p = \frac{\partial L}{\partial \dot{q}}$. The exact same cancellation occurs in the second equation.

To fully understand why this establishes equivalence, we must look at the theoretical implications of these two resulting derivatives.

The first result, $\frac{\partial H}{\partial p} = \dot{q}^T$ (or simply $\dot{q} = \frac{\partial H}{\partial p}$), is a kinematic relation. It inherently satisfies the definition of our conjugate momenta and ensures that our variable substitution via the Legendre transform remains consistent.

The second result yields $\frac{\partial H}{\partial q} = -\frac{\partial L}{\partial q}$. If we substitute the original second-order Euler-Lagrange equation, $\frac{\partial L}{\partial q} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right)$, and recall our momenta definition $p = \frac{\partial L}{\partial \dot{q}}$, the right side becomes precisely $-\dot{p}$. Therefore, we arrive at $\dot{p} = -\frac{\partial H}{\partial q}$.

Together, these steps demonstrate that the original, singular second-order Lagrangian system is perfectly decoupled into the symmetric pair of first-order Hamiltonian equations.

6 Symplectic Transformations

6.1 Oriented Area

To understand the geometric properties of Hamiltonian systems, we first consider a two-dimensional parallelogram in phase space spanned by two vectors $\xi = (\xi^p, \xi^q)^T$ and $\eta = (\eta^p, \eta^q)^T$.

In one dimension ($d = 1$), the oriented area of this parallelogram is simply the determinant of the coordinates:

$$\text{or.area}(P) = \det \begin{pmatrix} \xi^p & \eta^p \\ \xi^q & \eta^q \end{pmatrix} = \xi^p \eta^q - \xi^q \eta^p \quad (7)$$

For higher dimensions ($d > 1$), we generalize this by taking the sum of the oriented areas of the projections of the parallelogram onto the coordinate planes (p_i, q_i) . This is expressed using the bilinear mapping $\omega(\xi, \eta)$:

$$\omega(\xi, \eta) := \sum_{i=1}^d \det \begin{pmatrix} \xi_i^p & \eta_i^p \\ \xi_i^q & \eta_i^q \end{pmatrix} = \sum_{i=1}^d (\xi_i^p \eta_i^q - \xi_i^q \eta_i^p) \quad (8)$$

In matrix notation, this takes the elegant form $\omega(\xi, \eta) = \xi^T J \eta$, where $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ and I is the $d \times d$ identity matrix.

6.2 Defining Symplectic Maps

Definition 2.1. A linear mapping $A : \mathbb{R}^{2d} \rightarrow \mathbb{R}^{2d}$ is called *symplectic* if

$$A^T J A = J \quad (9)$$

or, equivalently, if $\omega(A\xi, A\eta) = \omega(\xi, \eta)$ for all $\xi, \eta \in \mathbb{R}^{2d}$.

This concept extends to non-linear maps by examining their local linear approximations (their Jacobians).

Definition 2.2. A differentiable map $g : U \rightarrow \mathbb{R}^{2d}$ (where $U \subset \mathbb{R}^{2d}$ is an open set) is called *symplectic* if the Jacobian matrix $g'(p, q)$ is everywhere symplectic, i.e., if

$$g'(p, q)^T J g'(p, q) = J \quad \text{or} \quad \omega(g'(p, q)\xi, g'(p, q)\eta) = \omega(\xi, \eta) \quad (10)$$

6.3 Geometric Interpretation on Manifolds

We can geometrically interpret this by considering a 2-dimensional sub-manifold M defined as the image $M = \psi(K)$ of a compact set $K \subset \mathbb{R}^2$, where $\psi(s, t)$ is a continuously differentiable parameterization.

If we divide the compact set K into infinitesimal compartments $ds dt$, the manifold M can be approximated as the union of small parallelograms spanned by the tangent vectors $\frac{\partial \psi}{\partial s}(s, t) ds$ and $\frac{\partial \psi}{\partial t}(s, t) dt$. Summing the oriented areas of the projections of all these infinitesimal parallelograms yields the total oriented area over the manifold:

$$\Omega(M) = \iint_K \omega \left(\frac{\partial \psi}{\partial s}(s, t), \frac{\partial \psi}{\partial t}(s, t) \right) ds dt \quad (11)$$

Lemma 2.3. *If the mapping $g : U \rightarrow \mathbb{R}^{2d}$ is symplectic on U , then it preserves the expression $\Omega(M)$, i.e.,*

$$\Omega(g(M)) = \Omega(M) \quad (12)$$

holds for all 2-dimensional manifolds M that can be represented as the image of a continuously differentiable function ψ .

Proof. The transformed manifold $g(M)$ can be parameterized by the composition $g \circ \psi$. Setting up the integral for the area of this new manifold, we have:

$$\Omega(g(M)) = \iint_K \omega \left(\frac{\partial(g \circ \psi)}{\partial s}(s, t), \frac{\partial(g \circ \psi)}{\partial t}(s, t) \right) ds dt \quad (13)$$

By the chain rule, the partial derivatives are $\frac{\partial}{\partial s}(g \circ \psi) = g'(\psi(s, t)) \frac{\partial \psi}{\partial s}$ (and identically for t). Because g is a symplectic transformation, its Jacobian g' preserves the bilinear map ω , meaning $\omega(g'\xi, g'\eta) = \omega(\xi, \eta)$. Therefore, the integrand simplifies entirely, yielding:

$$\Omega(g(M)) = \iint_K \omega \left(\frac{\partial \psi}{\partial s}(s, t), \frac{\partial \psi}{\partial t}(s, t) \right) ds dt = \Omega(M) \quad (14)$$

This confirms that any symplectic mapping (including non-linear ones) is fundamentally area-preserving.

7 Theorems on Symplectic Flow

Theorem 2.4 (Poincaré 1899). *Let $H(p, q)$ be a twice continuously differentiable function on $U \subset \mathbb{R}^{2d}$. Then, for each fixed t , the flow φ_t is a symplectic transformation wherever it is defined.*

Proof. Let $y = (p, q)$. We write the Hamiltonian system in the form $\dot{y} = J^{-1} \nabla H(y)$. The flow is given by $\varphi_t(y_0) = y(t)$.

We first derive the variational equation by differentiating the Jacobian of the flow, $\Psi = \frac{\partial \varphi_t}{\partial y_0}$, with respect to t :

$$\frac{d}{dt} \left(\frac{\partial \varphi_t}{\partial y_0} \right) = \frac{\partial}{\partial y_0} \left(\frac{d \varphi_t}{dt} \right) = \frac{\partial}{\partial y_0} (J^{-1} \nabla H(\varphi_t(y_0))) \quad (15)$$

Applying the chain rule yields:

$$\frac{d}{dt} \left(\frac{\partial \varphi_t}{\partial y_0} \right) = J^{-1} \nabla^2 H(\varphi_t(y_0)) \frac{\partial \varphi_t}{\partial y_0} \quad (16)$$

where $\nabla^2 H$ is the symmetric Hessian matrix of H . Thus, Ψ is a solution to the variational equation $\dot{\Psi} = J^{-1} \nabla^2 H(\varphi_t(y_0)) \Psi$.

To show the flow is symplectic, we differentiate the condition $\Psi^T J \Psi$ with respect to time:

$$\frac{d}{dt} (\Psi^T J \Psi) = \dot{\Psi}^T J \Psi + \Psi^T J \dot{\Psi} \quad (17)$$

Substituting our derived variational equation gives:

$$= (J^{-1} \nabla^2 H(\varphi_t(y_0)) \Psi)^T J \Psi + \Psi^T J (J^{-1} \nabla^2 H(\varphi_t(y_0)) \Psi) \quad (18)$$

Because $J^T = -J$ and $J^{-T} J = -I$, this simplifies to:

$$= \Psi^T \nabla^2 H(\varphi_t(y_0)) J^{-T} J \Psi + \Psi^T \nabla^2 H(\varphi_t(y_0)) \Psi = 0 \quad (19)$$

Since $\frac{d}{dt} (\Psi^T J \Psi) = 0$, the expression $\Psi^T J \Psi$ is constant for all t . At $t = 0$, φ_0 is the identity map, meaning $\Psi(0) = I$, which gives $I^T J I = J$. Therefore, $\left(\frac{\partial \varphi_t}{\partial y_0} \right)^T J \left(\frac{\partial \varphi_t}{\partial y_0} \right) = J$ for all t , proving the flow is symplectic.

Theorem 2.6. *Let $f : U \rightarrow \mathbb{R}^{2d}$ be continuously differentiable. Then, $\dot{y} = f(y)$ is locally Hamiltonian if and only if its flow $\varphi_t(y)$ is symplectic for all $y \in U$ and for all sufficiently small t .*

Proof. The necessity follows directly from Theorem 2.4. To prove sufficiency, we assume the flow φ_t is symplectic and must prove the local existence of a function $H(y)$ such that $f(y) = J^{-1} \nabla H(y)$.

We start by differentiating the symplecticity condition, $\left(\frac{\partial \varphi_t}{\partial y_0} \right)^T J \left(\frac{\partial \varphi_t}{\partial y_0} \right) = J$, with respect to t , knowing that $\frac{\partial \varphi_t}{\partial y_0}$ solves the variational equation $\dot{\Psi} = f'(\varphi_t(y_0)) \Psi$:

$$\frac{d}{dt} \left(\left(\frac{\partial \varphi_t}{\partial y_0} \right)^T J \left(\frac{\partial \varphi_t}{\partial y_0} \right) \right) = \left(\frac{\partial \varphi_t}{\partial y_0} \right)^T (f'(\varphi_t(y_0))^T J + J f'(\varphi_t(y_0))) \left(\frac{\partial \varphi_t}{\partial y_0} \right) = 0 \quad (20)$$

Putting $t = 0$, where $\varphi_0(y_0) = y_0$ and $\frac{\partial \varphi_0}{\partial y_0} = I$, it follows from $J = -J^T$ that $J f'(y_0)$ is a symmetric matrix for all y_0 .

To find H , let us assume $y_0 = 0$ and consider a ball around 0 contained in U . On this ball, we construct the function $H(y)$ as:

$$H(y) = \int_0^1 y^T J f(ty) dt + \text{Const} \quad (21)$$

Let $g(y) = Jf(y)$; since its Jacobian $g'(y) = Jf'(y)$ is symmetric, we have $\frac{\partial g_i}{\partial y_k} = \frac{\partial g_k}{\partial y_i}$. Differentiating $H(y)$ with respect to y_k yields:

$$\frac{\partial H}{\partial y_k}(y) = \int_0^1 \left(g_k(ty) + y^T \frac{\partial g}{\partial y_k}(ty)t \right) dt \quad (22)$$

Using the symmetry property, the integrand simplifies as an exact derivative:

$$\frac{\partial H}{\partial y_k}(y) = \int_0^1 \frac{d}{dt} (tg_k(ty)) dt = g_k(y) \quad (23)$$

This proves that $\nabla H(y) = g(y) = Jf(y)$. Rearranging gives $f(y) = J^{-1}\nabla H(y)$, concluding the proof that the system is locally Hamiltonian.

8 Symplecticity of the Symplectic Euler Method

To prove that the Symplectic Euler method is indeed symplectic, we examine the Jacobian of the transformation. The method is defined by the following step equations:

$$p_{n+1} = p_n - hH_q(p_{n+1}, q_n)$$

$$q_{n+1} = q_n + hH_p(p_{n+1}, q_n)$$

8.1 Step 1: Implicit Differentiation

We first differentiate both equations with respect to the initial states p_n and q_n .

Block A: Differentiating $p_{n+1} = p_n - hH_q(p_{n+1}, q_n)$

1. With respect to p_n :

$$\frac{\partial p_{n+1}}{\partial p_n} = I - hH_{qp}^T \frac{\partial p_{n+1}}{\partial p_n} - hH_{qq} \frac{\partial q_n}{\partial p_n}$$

Since $\frac{\partial q_n}{\partial p_n} = 0$, we get:

$$(I + hH_{qp}^T) \frac{\partial p_{n+1}}{\partial p_n} = I \quad (3)$$

2. With respect to q_n :

$$\frac{\partial p_{n+1}}{\partial q_n} = 0 - hH_{qp}^T \frac{\partial p_{n+1}}{\partial q_n} - hH_{qq} \frac{\partial q_n}{\partial q_n}$$

Since $\frac{\partial q_n}{\partial q_n} = I$, we get:

$$(I + hH_{qp}^T) \frac{\partial p_{n+1}}{\partial q_n} = -hH_{qq} \quad (4)$$

Block B: Differentiating $q_{n+1} = q_n + hH_p(p_{n+1}, q_n)$

1. With respect to p_n :

$$\frac{\partial q_{n+1}}{\partial p_n} = \frac{\partial q_n}{\partial p_n} + hH_{pp} \frac{\partial p_{n+1}}{\partial p_n} + hH_{qp} \frac{\partial q_n}{\partial p_n}$$

Since $\frac{\partial q_n}{\partial p_n} = 0$, we get:

$$-hH_{pp} \frac{\partial p_{n+1}}{\partial p_n} + \frac{\partial q_{n+1}}{\partial p_n} = 0 \quad (5)$$

2. With respect to q_n :

$$\frac{\partial q_{n+1}}{\partial q_n} = I + hH_{pp} \frac{\partial p_{n+1}}{\partial q_n} + hH_{qp} \frac{\partial q_n}{\partial q_n}$$

Since $\frac{\partial q_n}{\partial q_n} = I$, we get:

$$-hH_{pp} \frac{\partial p_{n+1}}{\partial q_n} + \frac{\partial q_{n+1}}{\partial q_n} = I + hH_{qp} \quad (6)$$

8.2 Step 2: The Compact Block Matrix Formulation

Combining equations (3), (4), (5), and (6) into a single block matrix system yields:

$$\begin{pmatrix} I + hH_{qp}^T & 0 \\ -hH_{pp} & I \end{pmatrix} \begin{pmatrix} \frac{\partial p_{n+1}}{\partial p_n} & \frac{\partial p_{n+1}}{\partial q_n} \\ \frac{\partial q_{n+1}}{\partial p_n} & \frac{\partial q_{n+1}}{\partial q_n} \end{pmatrix} = \begin{pmatrix} I & -hH_{qq} \\ 0 & I + hH_{qp} \end{pmatrix}$$

Let us define the helper matrices A and B :

$$A = \begin{pmatrix} I + hH_{qp}^T & 0 \\ -hH_{pp} & I \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} I & -hH_{qq} \\ 0 & I + hH_{qp} \end{pmatrix}$$

Thus, we can write the relation compactly as:

$$AG = B \implies G = A^{-1}B \quad (7)$$

8.3 Step 3: Proving Symplecticity using Transposes

In symplectic linear algebra, a matrix G is symplectic ($G^T J G = J$) if and only if its transpose is also symplectic ($G J G^T = J$). By using this equivalent definition, the proof collapses into three lines:

1. Substitute $G = A^{-1}B$ and $G^T = B^T A^{-T}$:

$$A^{-1}B J B^T A^{-T} = J$$

2. Multiply on the left by A :

$$A(A^{-1}B J B^T A^{-T}) = A J \implies B J B^T A^{-T} = A J$$

3. Multiply on the right by A^T :

$$(BJB^T A^{-T})A^T = AJA^T \implies BJB^T = AJA^T$$

This directly proves that showing $AJA^T = BJB^T$ is equivalent to showing the method is symplectic.

Part A: Calculating AJA^T Let $X = I + hH_{qp}^T$ and $Y = -hH_{pp}$ (where $Y^T = Y$ due to symmetry).

$$A = \begin{pmatrix} X & 0 \\ Y & I \end{pmatrix}$$

$$AJ = \begin{pmatrix} X & 0 \\ Y & I \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \begin{pmatrix} 0 & X \\ -I & Y \end{pmatrix}$$

$$AJA^T = \begin{pmatrix} 0 & X \\ -I & Y \end{pmatrix} \begin{pmatrix} X^T & Y^T \\ 0 & I \end{pmatrix} = \begin{pmatrix} 0 & X \\ -X^T & -Y^T + Y \end{pmatrix}$$

Since Y is symmetric, the bottom-right term $-Y^T + Y = 0$. Substituting X back in yields:

$$AJA^T = \begin{pmatrix} 0 & I + hH_{qp}^T \\ -(I + hH_{qp}) & 0 \end{pmatrix} \quad (9)$$

Part B: Calculating BJB^T Let $Z = -hH_{qq}$ (where $Z^T = Z$ due to symmetry) and $W = I + hH_{qp}$.

$$B = \begin{pmatrix} I & Z \\ 0 & W \end{pmatrix}$$

$$BJ = \begin{pmatrix} I & Z \\ 0 & W \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \begin{pmatrix} -Z & I \\ -W & 0 \end{pmatrix}$$

$$BJB^T = \begin{pmatrix} -Z & I \\ -W & 0 \end{pmatrix} \begin{pmatrix} I & 0 \\ Z^T & W^T \end{pmatrix} = \begin{pmatrix} -Z + Z^T & W^T \\ -W & 0 \end{pmatrix}$$

Since Z is symmetric, the top-left term $-Z + Z^T = 0$. Substituting W back in yields:

$$BJB^T = \begin{pmatrix} 0 & I + hH_{qp}^T \\ -(I + hH_{qp}) & 0 \end{pmatrix} \quad (10)$$

Comparing Equations (9) and (10), $AJA^T = BJB^T$. Therefore, the Symplectic Euler method is exactly symplectic.

9 Theorem 3.4 (The Implicit Midpoint Rule)

The implicit midpoint rule:

$$y_{n+1} = y_n + hJ^{-1}\nabla H\left(\frac{y_{n+1} + y_n}{2}\right) \quad (1)$$

is a symplectic method of order 2.

9.1 Proof of Symplecticity

Let the state vector be $y = (p, q)^T \in \mathbb{R}^{2d}$, and let J be the standard skew-symmetric structure matrix:

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$$

which satisfies the standard identities $J^T = -J$, $J^2 = -I$, and $J^{-1} = -J$.

We define the intermediate midpoint state as $y_{n+1/2} = \frac{y_{n+1} + y_n}{2}$ and the symmetric Hessian matrix of the Hamiltonian evaluated at this midpoint as:

$$S = \nabla^2 H(y_{n+1/2}) \quad \text{with} \quad S^T = S$$

Step 1: Implicit Differentiation To find the Jacobian matrix of the numerical step $G = \frac{\partial y_{n+1}}{\partial y_n}$, we differentiate both sides of Equation (1) with respect to the initial state y_n :

$$\frac{\partial y_{n+1}}{\partial y_n} = \frac{\partial y_n}{\partial y_n} + hJ^{-1} \frac{\partial}{\partial y_n} [\nabla H(y_{n+1/2})]$$

Applying the multivariable chain rule to the gradient term yields:

$$\frac{\partial y_{n+1}}{\partial y_n} = I + hJ^{-1} \left(\nabla^2 H(y_{n+1/2}) \cdot \frac{\partial y_{n+1/2}}{\partial y_n} \right)$$

$$G = I + hJ^{-1}S \left[\frac{1}{2} \left(\frac{\partial y_{n+1}}{\partial y_n} + \frac{\partial y_n}{\partial y_n} \right) \right]$$

$$G = I + \frac{h}{2}J^{-1}S(G + I)$$

Step 2: Isolating the Jacobian G Grouping the terms containing G on the left-hand side gives:

$$G - \frac{h}{2}J^{-1}SG = I + \frac{h}{2}J^{-1}S$$

$$\left(I - \frac{h}{2}J^{-1}S \right) G = I + \frac{h}{2}J^{-1}S \quad (2)$$

Let us define the helper matrices A and B as:

$$A = I - \frac{h}{2}J^{-1}S \quad \text{and} \quad B = I + \frac{h}{2}J^{-1}S$$

This simplifies our algebraic relation to $AG = B$, which means the Jacobian is given by $G = A^{-1}B$.

Step 3: Proving $G^T JG = J$ To show that the Jacobian G is symplectic, we must prove that $G^T JG = J$. Substituting $G = A^{-1}B$, we can rewrite this condition as:

$$(A^{-1}B)^T J(A^{-1}B) = J \implies B^T A^{-T} J A^{-1} B = J$$

Multiplying by A^T on the left and A on the right eliminates the inverse matrices without making any illegal commutation assumptions:

$$B^T J B = A^T J A \quad (3)$$

Let us expand both sides of Equation (3) to prove they are identical:

1. Expanding the Left-Hand Side ($A^T J A$): Using the property $J^{-1} = -J$, we write $A = I + \frac{h}{2}JS$. Its transpose is:

$$A^T = \left(I + \frac{h}{2}JS \right)^T = I + \frac{h}{2}S^T J^T = I - \frac{h}{2}SJ$$

Now, we compute the product $A^T J A$:

$$A^T J A = \left(I - \frac{h}{2}SJ \right) J \left(I + \frac{h}{2}JS \right)$$

$$A^T J A = \left(J - \frac{h}{2}SJ^2 \right) \left(I + \frac{h}{2}JS \right)$$

Since $J^2 = -I$, this simplifies to:

$$A^T J A = \left(J + \frac{h}{2}S \right) \left(I + \frac{h}{2}JS \right)$$

$$A^T J A = J + \frac{h}{2}J^2 S + \frac{h}{2}S + \frac{h^2}{4}SJS$$

Using $J^2 = -I$ once more:

$$A^T J A = J - \frac{h}{2}S + \frac{h}{2}S + \frac{h^2}{4}SJS = J + \frac{h^2}{4}SJS \quad (4)$$

2. Expanding the Right-Hand Side ($B^T J B$): Similarly, we write $B = I - \frac{h}{2}JS$. Its transpose is:

$$B^T = \left(I - \frac{h}{2}JS \right)^T = I - \frac{h}{2}S^T J^T = I + \frac{h}{2}SJ$$

Computing $B^T J B$:

$$\begin{aligned}
B^T J B &= \left(I + \frac{h}{2} S J \right) J \left(I - \frac{h}{2} J S \right) \\
B^T J B &= \left(J + \frac{h}{2} S J^2 \right) \left(I - \frac{h}{2} J S \right) = \left(J - \frac{h}{2} S \right) \left(I - \frac{h}{2} J S \right) \\
B^T J B &= J - \frac{h}{2} J^2 S - \frac{h}{2} S + \frac{h^2}{4} S J S
\end{aligned}$$

Using $J^2 = -I$:

$$B^T J B = J + \frac{h}{2} S - \frac{h}{2} S + \frac{h^2}{4} S J S = J + \frac{h^2}{4} S J S \quad (5)$$

Conclusion Comparing Equation (4) and Equation (5), we have proven rigorously that:

$$A^T J A = B^T J B = J + \frac{h^2}{4} S J S$$

Because this identity holds without requiring any commutation of S and J , the Jacobian $G = A^{-1}B$ is mathematically guaranteed to satisfy the symplecticity condition $G^T J G = J$ exactly. This completes the proof of Theorem 3.4.